

Muhammad Aswad Yousaf

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EDUCATION

National University of Modern Languages (NUML)

Bachelor of Science in Computer Science

Islamabad, Pakistan

Sep. 2022 – Jun. 2026

EXPERIENCE

Senior Web Developer

2022 – Present

Freelance Work

Pakistan

- Designed and developed full-stack web applications for clients across various industries
- Built responsive, production-ready websites using React.js, Django, and Python
- Collaborated with clients to gather requirements, deliver iterative prototypes, and deploy solutions
- Maintained and optimized existing codebases, improving performance and user experience

AI/ML Engineer

2024 – Present

Freelance Work

Pakistan

- Fine-tuned large language models on custom datasets to improve domain-specific performance
- Implemented Retrieval-Augmented Generation (RAG) pipelines for intelligent document Q&A systems
- Integrated AI/ML models into web applications via REST APIs built with Django
- Experimented with prompt engineering and embedding strategies to optimize model outputs

PROJECTS

C++-in-Urdu | C++

2023 – Present

- Created an open-source educational resource teaching C++ programming concepts in the Urdu language
- Lowered the barrier to entry for Urdu-speaking students learning systems programming
- Covered core topics including data structures, algorithms, OOP, and memory management

WhatsApp Bot | Python, Django

2024 – Present

- Developed an automated WhatsApp bot using Python and Django for message handling and routing
- Integrated webhook-based communication to process and respond to user queries in real time
- Designed a modular command system enabling easy extension of bot functionality

E-Commerce Platform | Python, Django, TypeScript

2024

- Built a full-stack e-commerce platform with a Django REST Framework backend backed by MySQL
- Implemented modular Django apps for products, cart, and orders with serialized REST API endpoints
- Developed a TypeScript frontend consuming the REST API with cart management and checkout flow

Real-Time Chat Application | Java, Spring Boot, WebSocket

2024

- Developed a real-time multi-user chat application using Spring Boot and the STOMP WebSocket protocol
- Configured a WebSocket message broker and chat controller to handle broadcast and targeted messaging
- Built an interactive HTML/CSS/JavaScript frontend for live conversations without page reloads

YouTube Content Downloader | Python, Django, yt-dlp

2024

- Built a Django backend enabling users to search and download YouTube audio and video content
- Integrated yt-dlp for reliable media extraction, format selection, and local file management
- Deployed to Heroku with a Procfile-based production configuration for cloud hosting

ACHIEVEMENTS & HACKATHONS

1st Position – UMT Surge Hackathon

2024 & 2025

University of Management and Technology

1st Position – ITU Hackathon

2025

Information Technology University

1st Position – Riphah RC3

2025

Riphah International University

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Frameworks: Django, Spring Boot, React.js, Next.js

AI/ML: Model Fine-tuning, Retrieval-Augmented Generation (RAG), LLMs

Databases: MySQL, SQLite

Game Engines: Unity (1 yr), Unreal Engine 5 (1.5 yrs)

Developer Tools: Git, Docker, Heroku, TravisCI, Google Cloud Platform, VS Code, PyCharm, IntelliJ, Postman